Efficient Exploration by HyperDQN

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Abstract

Exploration-exploitation trade-off is crucial to sample-efficient reinforcement learning. In this paper, we present a salable exploration method called *HyperDQN*, which

- Extends a probabilistic meta-model.
- Captures the **epistemic uncertainty** of the Q-value function over the **parameter space**.
- Samples **randomized Q-value functions**, which generates exploratory action sequences for deep exploration.

Motivation

- Thompson sampling (TS) methods balance between exploration and exploitation by sampling plausible actions from the posterior distribution.
- It is difficult to apply TS to complicated reinforcement learning problems since we cannot obtain the exact posterior distribution.
- Instead, we turn to a promising approach called randomized least-square value iteration (RLSVI) that is compatible with linearly parameterized value functions.
- We extend the idea of RLSVI to the non-linear function (neural network) approximation setting.

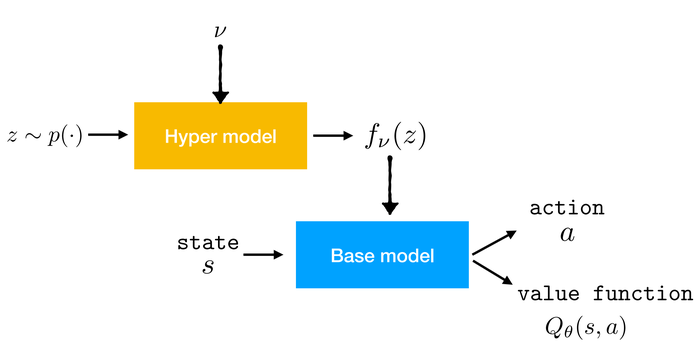


Figure: Illustration for HyperDQN.

Capturing Epistemic Uncertainty

- Hyper model maps a random vector z from the "prior" to the base model parameter θ from the "posterior".
- Each random vector z corresponds to one specific base model; as such, the hyper model can manage "infinite" base models.
- Multiple invocations yield different value functions $Q_\theta(s,a)$ and the induced greedy policies $\pi_\theta(a|s)$.

Randomized Temporal Difference

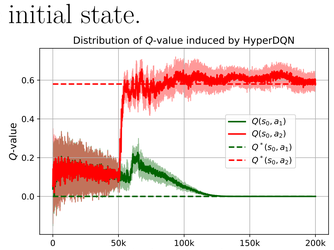
We update parameters of hyper model by randomized temporal difference learning to mimic RLSVI:

$$\min_{\nu} \sum_{\substack{(s,a,r,\xi,s')\in\mathcal{D},\\ z\in\mathcal{Z}}} \left(\sigma_\omega z^\top \xi + r + \gamma \max_{a'} Q_{f_{\bar{\nu}}(z)}(s',a') - Q_{f_\nu(z)}(s,a)\right)^2 + \frac{\sigma_w^2}{\sigma_p^2}\left\|\nu-\nu_0\right\|^2.$$

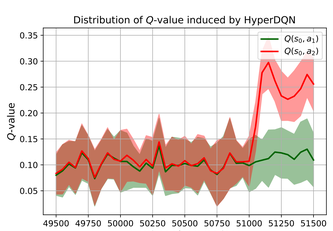
- The first part amounts to a standard temporal difference learning with an additional noise on the target due to "maximum likelihood estimation".
- The second part is intimately related to the "prior regularization".
- Overall, the base model θ obtained by $f_\nu(z)$ is treated as random variables so that the hyper model can capture the epistemic uncertainty from observed data.

Visualization

We investigate the distribution of Q-value induced by HyperDQN at the initial state.



(a) The distribution of Q-value.

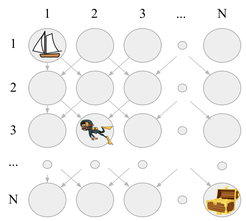


(b) A zoom-in of (a).

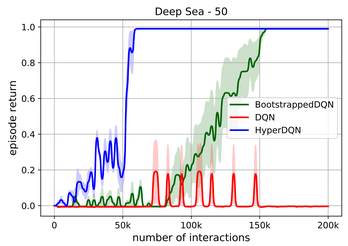
- The optimal action becomes dominant in sampled randomized Q-value functions after $50k$ interactions.

Empirical Performance

We validate the proposed method on two hard exploration tasks with sparse reward and/or long horizon.

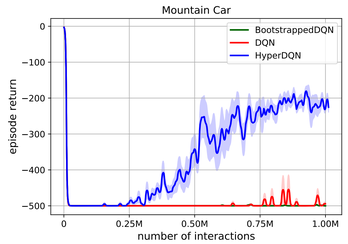


(a) Deep sea.



(b) Performance.

(a) Mountain Car.



(b) Performance.

- HyperDQN can achieve *deep exploration* and outperform baselines.

Conclusion

- Highlight deep exploration by randomized value functions generated by HyperDQN for efficient exploration.
- Extend to continuous control tasks in the future work.

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